



Unit 1: Matrices

Matrices, Operations on Matrices,
Determinant, System of Equations

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1 Introduction to Matrices

1.1 What is a Matrix?

A **matrix** is a rectangular arrangement of numbers (or expressions) in rows and columns. It provides a compact, powerful, and systematic way to organize data, solve systems of equations, and model real-world problems.

Definition: A matrix of order $m \times n$ is a rectangular array of mn numbers arranged in m rows and n columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The numbers a_{ij} are called **elements** or **entries**. The element a_{ij} lies in the i -th row and j -th column.

1.2 Why Do We Need Matrices? (Real-Life Examples)

Before learning the formal rules, let us look at several concrete situations where matrices naturally appear.

Example 1 (Class Marks): The marks obtained by 4 students in 3 subjects are:

	Maths	Physics	Chemistry
Aarav	85	78	92
Diya	90	82	88
Kabir	76	91	80
Meera	88	75	95

This entire table of numbers can be written as a matrix of order 4×3 :

$$M = \begin{pmatrix} 85 & 78 & 92 \\ 90 & 82 & 88 \\ 76 & 91 & 80 \\ 88 & 75 & 95 \end{pmatrix}$$

Here $a_{21} = 90$ means Diya scored 90 in Maths.

Example 2 (Pixel Grid of a Photograph): A small 3×3 grayscale image is stored as:

$$P = \begin{pmatrix} 255 & 128 & 64 \\ 200 & 30 & 180 \\ 90 & 220 & 10 \end{pmatrix}$$

Each entry is a pixel intensity between 0 (black) and 255 (white). Image processing operations (blurring, sharpening) are matrix operations.

Example 3 (Distance Between Cities): The road distances (in km) between four cities A, B, C, D:

$$D = \begin{pmatrix} 0 & 120 & 340 & 210 \\ 120 & 0 & 280 & 190 \\ 340 & 280 & 0 & 410 \\ 210 & 190 & 410 & 0 \end{pmatrix}$$

Notice that D is symmetric: $d_{ij} = d_{ji}$.

Example 4 (Sales Data): A shop sells two products P1 and P2 in three months (Jan, Feb, Mar). The sales data can be stored as:

$$S = \begin{pmatrix} 150 & 200 \\ 175 & 180 \\ 210 & 160 \end{pmatrix}$$

Forensic Application: In a forensic lab, the concentration (mg/L) of three toxins (T1, T2, T3) found in four blood samples (S1–S4) can be stored as:

$$C = \begin{pmatrix} 2.1 & 0.5 & 1.2 \\ 1.8 & 0.9 & 0.7 \\ 0.5 & 2.1 & 1.5 \\ 1.1 & 0.4 & 2.0 \end{pmatrix}$$

Row 3 shows that sample S3 has 0.5 mg/L of T1, 2.1 mg/L of T2, and 1.5 mg/L of T3.

1.3 Notation and Terminology

- A matrix is usually denoted by capital letters: $A, B, C, \dots$
- The **order** (or size) of a matrix with m rows and n columns is $m \times n$.
- If $m = n$, the matrix is called a **square matrix** of order n .
- The set of all $m \times n$ matrices with real entries is denoted by $M_{m \times n}(\mathbb{R})$.

Example 5: Consider

$$A = \begin{pmatrix} 2 & 5 & -1 \\ 0 & 3 & 4 \end{pmatrix}$$

- Order of A is 2×3 .
- $a_{11} = 2$, $a_{12} = 5$, $a_{13} = -1$.

- $a_{21} = 0, a_{22} = 3, a_{23} = 4.$

2 Types of Matrices

Matrices come in many special forms. Identifying the type helps us understand the matrix's structure and often simplifies computations.

2.1 Zero Matrix and Identity Matrix (First and Foremost)

Before listing other types, we introduce the two most fundamental matrices that play the role of “0” and “1” in matrix algebra.

Zero (Null) Matrix: A matrix in which **every** entry is zero is called the zero matrix. It is denoted by O or $O_{m \times n}$. It plays the same role as the number 0 in ordinary arithmetic.

$$O_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad O_{3 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Key property: $A + O = A$ and $A \cdot O = O \cdot A = O$ (when dimensions permit).

Identity Matrix: A square matrix in which all **diagonal entries are 1** and all **off-diagonal entries are 0** is called the identity matrix. It is denoted by I_n (where n is the order). It plays the same role as the number 1 in ordinary arithmetic.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Key property: $AI_n = A$ and $I_n A = A$ (when dimensions permit).

2.2 Complete List of Matrix Types

1. **Row Matrix:** Has only one row. Example: $(1 \ 3 \ 5)$ (order 1×3).
2. **Column Matrix:** Has only one column. Example: $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ (order 3×1).
3. **Rectangular Matrix:** $m \neq n$. Example: order 2×3 .
4. **Square Matrix:** $m = n$. Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (order 2×2).
5. **Diagonal Matrix:** A square matrix where all off-diagonal entries are zero:

$$D = \begin{pmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{pmatrix}$$

6. **Scalar Matrix:** A diagonal matrix where all diagonal entries are equal:

$$S = \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

For example, $3I_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is a scalar matrix.

7. **Identity Matrix:** A scalar matrix with $k = 1$. Denoted I_n . (See definition above.)

8. **Zero (Null) Matrix:** All entries are zero. (See definition above.)

9. **Symmetric Matrix:** A square matrix with $A^T = A$, i.e., $a_{ij} = a_{ji}$. Example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 6 & 9 \end{pmatrix}$$

10. **Skew-Symmetric Matrix:** A square matrix with $A^T = -A$, i.e., $a_{ij} = -a_{ji}$ and $a_{ii} = 0$. Example:

$$\begin{pmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{pmatrix}$$

11. **Upper Triangular Matrix:** Square matrix where all entries below the main diagonal are zero:

$$U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix}$$

12. **Lower Triangular Matrix:** Square matrix where all entries above the main diagonal are zero:

$$L = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Forensic Application: Fingerprints of 5 suspects, each represented by 4 ridge-count features (left thumb, right thumb, left index, right index). The data forms a 5×4 matrix. If we normalize and consider relationships between features, a **symmetric matrix** (of correlations) of order 4×4 arises.

3 Operations on Matrices

3.1 Equality of Matrices

Two matrices A and B are **equal** ($A = B$) if and only if:

1. They have the same order (same number of rows and columns).
2. Corresponding elements are equal: $a_{ij} = b_{ij}$ for all i, j .

Example 7: Find x, y if

$$\begin{pmatrix} x & 2 \\ 3 & y-1 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$$

Solution: Equating corresponding elements:

$$x = 5$$

$$y - 1 = 4 \implies y = 5$$

(x and y are uniquely determined.)

3.2 Addition and Subtraction of Matrices

Rule: Two matrices can be added/subtracted **only if** they have the same order. The result is obtained by adding/subtracting corresponding elements:

$$(A \pm B)_{ij} = a_{ij} \pm b_{ij}$$

Example 8: If

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & -1 \\ 0 & 2 \end{pmatrix}$$

then

$$A + B = \begin{pmatrix} 6 & 1 \\ 3 & 6 \end{pmatrix}, \quad A - B = \begin{pmatrix} -4 & 3 \\ 3 & 2 \end{pmatrix}$$

Properties of Matrix Addition:

1. **Commutative:** $A + B = B + A$.
2. **Associative:** $(A + B) + C = A + (B + C)$.
3. **Additive Identity:** $A + O = A$.
4. **Additive Inverse:** $A + (-A) = O$.

3.3 Scalar Multiplication

Rule: To multiply a matrix A by a scalar k , multiply every element of A by k :

$$(kA)_{ij} = k \cdot a_{ij}$$

Example 9: If $A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$ and $k = 4$, then

$$4A = \begin{pmatrix} 8 & -4 \\ 0 & 12 \end{pmatrix}$$

Properties of Scalar Multiplication:

1. $k(A + B) = kA + kB$
2. $(k + l)A = kA + lA$
3. $k(lA) = (kl)A$
4. $1 \cdot A = A, 0 \cdot A = O$

3.4 Matrix Multiplication (Detailed and Visualized)

Matrix multiplication is the most important—and most challenging—operation in this course. Unlike addition, it is **not** done element-by-element.

3.4.1 The Rule and the Condition

Rule: If A is of order $m \times n$ and B is of order $n \times p$, then the product AB is defined and has order $m \times p$. Its (i, j) -th entry is:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

This is the **dot product** of the i -th row of A and the j -th column of B .

Dimension check (the “book-keeping” rule):

$$\underbrace{A}_{m \times n} \cdot \underbrace{B}_{n \times p} = \underbrace{AB}_{m \times p}$$

The **inner dimensions must match** (columns of A = rows of B). The **outer dimensions** give the answer.

3.4.2 Visualization of Matrix Multiplication

Let us visualize the process step by step.

Example 11 (Visual Walkthrough): Let

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

We compute AB :

Step 1: The $(1, 1)$ -entry of AB = Row 1 of A · Column 1 of B :

$$\begin{array}{l} \text{Row 1 of } A : \boxed{a_{11} \ a_{12}} \\ \text{Column 1 of } B : \boxed{\begin{matrix} b_{11} \\ b_{21} \end{matrix}} \end{array} \implies a_{11}b_{11} + a_{12}b_{21}$$

Step 2: The $(1, 2)$ -entry of AB = Row 1 of A · Column 2 of B :

$$\boxed{a_{11} \ a_{12}} \cdot \boxed{\begin{matrix} b_{12} \\ b_{22} \end{matrix}} = a_{11}b_{12} + a_{12}b_{22}$$

Step 3: The $(2, 1)$ -entry of AB = Row 2 of A · Column 1 of B :

$$\begin{bmatrix} a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = a_{21}b_{11} + a_{22}b_{21}$$

Step 4: The $(2, 2)$ -entry of AB = Row 2 of A · Column 2 of B :

$$a_{21}b_{12} + a_{22}b_{22}$$

Therefore:

$$AB = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

“Row-Column Rule”: To get the (i, j) -entry of AB , take the i -th row of A , place it next to the j -th column of B , and multiply-and-add term by term.

$$\text{Row } i \text{ of } A : (a_{i1}, a_{i2}, \dots, a_{in})$$

$$\text{Column } j \text{ of } B : \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}$$

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

3.4.3 Fully Worked Numerical Examples

Example 12: Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

Compute AB :

$$\begin{aligned} AB &= \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} \\ &= \begin{pmatrix} 5 + 14 & 6 + 16 \\ 15 + 28 & 18 + 32 \end{pmatrix} \\ &= \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \end{aligned}$$

Compute BA :

$$\begin{aligned} BA &= \begin{pmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 5 + 18 & 10 + 24 \\ 7 + 24 & 14 + 32 \end{pmatrix} = \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix} \end{aligned}$$

Notice $AB \neq BA$. This is called **non-commutativity** of matrix multiplication.

Example 13 (Non-Square): If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}_{2 \times 3}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}_{3 \times 2}$$

then AB is defined (inner dimensions: $3 = 3$) and has order 2×2 :

$$AB = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 + 3 \cdot 1 \\ 4 \cdot 1 + 5 \cdot 0 + 6 \cdot 1 & 4 \cdot 0 + 5 \cdot 1 + 6 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 10 & 11 \end{pmatrix}$$

But BA is 3×3 . So $BA \neq AB$ (not even the same size).

3.4.4 Two Examples Showing $AB \neq BA$

Example 14 $AB \neq BA$ – – – *Version1*: Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Then

$$AB = \begin{pmatrix} 0 \cdot 0 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$BA = \begin{pmatrix} 0 \cdot 0 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 \\ 1 \cdot 0 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Clearly $AB \neq BA$.

Example 15 $AB \neq BA$ – – – *Version2*: Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$$

Then:

$$AB = \begin{pmatrix} 2 \cdot 1 + 1 \cdot 4 & 2 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 1 + 3 \cdot 4 & 1 \cdot 0 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 13 & 3 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 \cdot 2 + 0 \cdot 1 & 1 \cdot 1 + 0 \cdot 3 \\ 4 \cdot 2 + 1 \cdot 1 & 4 \cdot 1 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 9 & 7 \end{pmatrix}$$

Thus $AB = \begin{pmatrix} 6 & 1 \\ 13 & 3 \end{pmatrix} \neq \begin{pmatrix} 2 & 1 \\ 9 & 7 \end{pmatrix} = BA$.

3.4.5 Properties of Matrix Multiplication

1. **Associative:** $(AB)C = A(BC)$ (when dimensions permit).
2. **Distributive:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
3. **Identity:** $AI = IA = A$.

4. **Not Commutative:** $AB \neq BA$ in general (Examples 14, 15).
5. **Zero Product:** $AB = O$ does not imply $A = O$ or $B = O$.

Example 16 (Zero Product): Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Then

$$AB = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O$$

Neither A nor B is zero, yet $AB = O$.

3.4.6 Importance of Scalar Multiplication with Diagonal Matrices

Multiplying by a diagonal matrix is a very elegant operation. Let us see why.

Key Result: If $D = \text{diag}(d_1, d_2, \dots, d_n)$ is a diagonal matrix and A is an $n \times n$ matrix, then:

- DA scales each row of A by d_i (the i -th row is multiplied by d_i).
- AD scales each column of A by d_j (the j -th column is multiplied by d_j).

Example 10: Let

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Compute DA :

$$DA = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 12 & 15 & 18 \\ 35 & 40 & 45 \end{pmatrix}$$

Observe: row 1 is multiplied by 2, row 2 by 3, and row 3 by 5.

Now compute AD :

$$AD = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 6 & 15 \\ 8 & 15 & 30 \\ 14 & 24 & 45 \end{pmatrix}$$

Observe: column 1 is multiplied by 2, column 2 by 3, and column 3 by 5.

4 Polynomial Expressions in a Matrix

4.1 Motivation: Why Evaluate Polynomials in A ?

We often encounter expressions such as

$$A^2 + 5A - 2I, \quad A^3 - 4A^2 + 3A - I,$$

where A is a given square matrix and I is the identity matrix of the same order. Computing these directly is straightforward but tedious for higher powers.

4.2 Worked Examples

Example 1 (2×2): Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Compute $A^2 - 5A - 2I$.

Solution:

First compute A^2 :

$$A^2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}.$$

Now $5A = \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix}$, and $2I = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

Thus:

$$A^2 - 5A - 2I = \begin{pmatrix} 7-5-2 & 10-10-0 \\ 15-15-0 & 22-20-2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O.$$

Example 2 (3×3): Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Verify $A^3 - 4A^2 + 5A - 2I = O$.

Solution:

Now compute powers of A :

$$A^2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

Now:

$$A^3 - 4A^2 + 5A - 2I = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 8 \end{pmatrix} - \begin{pmatrix} 4 & 8 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 16 \end{pmatrix} + \begin{pmatrix} 5 & 5 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 10 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Add entrywise:

$$= \begin{pmatrix} 1 - 4 + 5 - 2 & 3 - 8 + 5 - 0 & 0 \\ 0 & 1 - 4 + 5 - 2 & 0 \\ 0 & 0 & 8 - 16 + 10 - 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = O. \checkmark$$

Forensic Application: In modelling the spread of a contaminant across a crime scene grid, the transition matrix A satisfies a characteristic equation. Polynomial expressions like $A^2 - 5A + 3I$ arise in stability analysis (e.g., checking whether a Markov chain settles down).

5 Determinants

5.1 Definition

The **determinant** is a scalar value associated with a square matrix. It is denoted by $\det(A)$ or $|A|$.

5.1.1 Determinant of a 2×2 Matrix

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det(A) = |A| = ad - bc$$

Example 18: If $A = \begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix}$, then

$$|A| = (3)(5) - (4)(2) = 15 - 8 = 7$$

5.1.2 Determinant of a 3×3 Matrix

For

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

expanding along the first row:

$$|A| = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Example 19: Compute $|A|$ for

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix}$$

$$\begin{aligned} |A| &= 1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 10 \end{vmatrix} - 2 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 10 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 1(50 - 48) - 2(40 - 42) + 3(32 - 35) \\ &= 1(2) - 2(-2) + 3(-3) = 2 + 4 - 9 = -3 \end{aligned}$$

5.2 Properties of Determinants

1. If any row/column is entirely zero, $|A| = 0$.
2. If two rows/columns are identical, $|A| = 0$.
3. Interchanging two rows/columns changes the sign of $|A|$.
4. Multiplying a row by scalar k multiplies $|A|$ by k .
5. Adding a multiple of one row to another leaves $|A|$ unchanged.
6. $|AB| = |A| \cdot |B|$.
7. $|A^T| = |A|$.
8. For triangular A , $|A| = \text{product of diagonal entries}$.
9. $|kA| = k^n |A|$ for $n \times n$ matrix.
10. A is invertible iff $|A| \neq 0$.

6 System of Linear Equations and Cramer's Rule

6.1 Introduction

A system of m linear equations in n unknowns has the form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

In matrix form:

$$\underbrace{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}}_X = \underbrace{\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}}_B$$

i.e., $AX = B$.

6.2 Types of Solutions

A linear system can have:

1. **Unique solution** (exactly one).
2. **Infinitely many solutions**.
3. **No solution** (inconsistent).

For a **square system** (n equations, n unknowns), with $|A| \neq 0$:

- The system has a **unique solution**.
- The solution can be found using **Cramer's rule** (below).

If $|A| = 0$, the system has either no solution or infinitely many.

6.3 Cramer's Rule (The Focus Method)

Cramer's Rule: For $AX = B$ with A an $n \times n$ matrix and $|A| \neq 0$, the unique solution is

$$x_1 = \frac{|A_1|}{|A|}, \quad x_2 = \frac{|A_2|}{|A|}, \quad \dots, \quad x_n = \frac{|A_n|}{|A|}$$

where A_i is the matrix obtained by replacing the i -th **column** of A with the constants column B .

6.3.1 Worked Examples

Example 21 (2×2): Solve

$$\begin{aligned} 2x + y &= 7 \\ 5x + 3y &= 18 \end{aligned}$$

Solution:

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}, \quad |A| = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1 \neq 0$$

Replace column 1 with $B = \begin{pmatrix} 7 \\ 18 \end{pmatrix}$:

$$A_1 = \begin{pmatrix} 7 & 1 \\ 18 & 3 \end{pmatrix}, \quad |A_1| = 21 - 18 = 3$$

Replace column 2 with B :

$$A_2 = \begin{pmatrix} 2 & 7 \\ 5 & 18 \end{pmatrix}, \quad |A_2| = 36 - 35 = 1$$

$$x = \frac{|A_1|}{|A|} = \frac{3}{1} = 3, \quad y = \frac{|A_2|}{|A|} = \frac{1}{1} = 1$$

So $x = 3, y = 1$. Verification: $2(3) + 1 = 7 \checkmark$.

Example 22 (3×3): Solve

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y - z = 2$$

Solution:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

Compute $|A|$ by first-row expansion:

$$\begin{aligned} |A| &= 1 \cdot \begin{vmatrix} -1 & 1 \\ 2 & -1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} \\ &= 1(1 - 2) - 1(-2 - 1) + 1(4 + 1) \\ &= -1 + 3 + 5 = 7 \neq 0 \end{aligned}$$

For x : Replace column 1 with $B = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}$:

$$A_1 = \begin{pmatrix} 6 & 1 & 1 \\ 3 & -1 & 1 \\ 2 & 2 & -1 \end{pmatrix}$$

$$|A_1| = 6(1 - 2) - 1(-3 - 2) + 1(6 + 2) = -6 + 5 + 8 = 7$$

$$x = \frac{|A_1|}{|A|} = \frac{7}{7} = 1$$

For y : Replace column 2 with B :

$$A_2 = \begin{pmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

$$|A_2| = 1(-3 - 2) - 6(-2 - 1) + 1(4 - 3) = -5 + 18 + 1 = 14$$

$$y = \frac{14}{7} = 2$$

For z : Replace column 3 with B :

$$A_3 = \begin{pmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 1 & 2 & 2 \end{pmatrix}$$

$$|A_3| = 1(-2 - 6) - 1(4 - 3) + 6(4 + 1) = -8 - 1 + 30 = 21$$

$$z = \frac{21}{7} = 3$$

Solution: $x = 1, y = 2, z = 3$.

Forensic Application of Cramer's Rule: Suppose a forensic chemist mixes three standard solutions to produce a calibration mixture. The volumes x, y, z (in mL) of the three standards satisfy:

$$x + 2y + 3z = 20$$

$$2x + y + z = 14$$

$$3x - y + 2z = 15$$

Using Cramer's rule, the chemist can find the exact volumes needed. This kind of system appears regularly in analytical toxicology when preparing known-concentration standards for instrumental calibration.

6.4 Homogeneous Systems

A system $AX = O$ (where $B = O$) is **homogeneous**. It always has the trivial solution $X = O$.

Key Result:

- If $|A| \neq 0$, the only solution is the trivial solution $x_1 = x_2 = \dots = x_n = 0$.
- If $|A| = 0$, there are infinitely many non-trivial solutions.

7 Applications in Forensic Science

Application 1: Forensic Toxicology (Mixture Analysis). Suppose a blood sample is a mixture of three drugs with unknown concentrations c_1, c_2, c_3 . The lab measures three different absorbances at three wavelengths. The measurement model gives:

$$2.1c_1 + 0.5c_2 + 1.2c_3 = 8.3$$

$$1.8c_1 + 0.9c_2 + 0.7c_3 = 6.4$$

$$0.5c_1 + 2.1c_2 + 1.5c_3 = 7.8$$

Using Cramer's rule, we can solve for c_1, c_2, c_3 .

Application 2: Fingerprint Matching. Fingerprint features (ridge counts between reference points) are stored as vectors. Matching a test print against a database requires computing distances between vectors – essentially row-column dot products from matrix multiplication.

Application 3: Crime Scene Reconstruction. Physical evidence (bullet tra-

jectory, blood spatter angles, witness statements) generates a system of linear equations. Cramer's rule provides a direct algebraic solution.

Application 4: DNA Profile Probabilities. In forensic genetics, allele frequency data are stored in matrices. Population statistics are computed using matrix products – a direct application of matrix multiplication.

Application 5: Image Enhancement. CCTV footage often has low contrast. A diagonal scaling matrix D is applied to pixel intensity matrices. This uses the fact that DA scales each row of A by d_i – an elegant result of diagonal matrices.

8 Practice Questions

1. If $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 5 \\ -2 & 1 \end{pmatrix}$, compute $A + B$, $A - B$, and $3A - 2B$.

2. Find AB and BA for

$$A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 0 \\ 1 & 5 \end{pmatrix}$$

Is $AB = BA$? Explain with a reason.

3. If $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, compute DA and AD . Describe in words what happened to the rows and columns of A .

4. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute:

(a) A^2

(b) $A^2 - 5A - 2I$

5. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Compute:

(a) $A^2 + 5A - 2I$

(b) Verify the equation $A^2 - 5A + 5I = O$.

6. Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Verify $A^3 - 6A^2 + 12A - 8I = O$.

7. Write $A = \begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

8. Decompose $A = \begin{pmatrix} 1 & 4 & -2 \\ 6 & 0 & 3 \\ 5 & -1 & 2 \end{pmatrix}$ into symmetric and skew-symmetric parts.

9. Given that $A = \begin{pmatrix} 0 & 5 \\ -5 & 0 \end{pmatrix}$. Calculate the matrix $A^2 + 5A - 3I$.

10. Given that

$$A = \begin{pmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 2 & 6 \end{pmatrix}$$

. Calculate $A^2 - 2A + 5I$.

11. If $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$, find S and K and verify $S^T = S$, $K^T = -K$.

12. Compute the determinant of

$$A = \begin{pmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 2 & 6 \end{pmatrix}$$

using expansion along the second row.

13. Compute the determinant of

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

using the row or column with the most zeros.

14. Use Cramer's rule to solve:

$$2x + 3y = 13$$

$$4x - y = 5$$

15. Use Cramer's rule to solve:

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y - z = 2$$

16. A forensic chemist prepares a calibration standard by mixing three stock solutions with concentrations a, b, c (in % v/v). The mixing conditions are:

$$a + b + c = 100$$

$$2a - b = 10$$

$$a - 2c = -40$$

Use Cramer's rule to find a, b, c .

17. A forensic lab tests three blood samples for three substances. The concentration matrix (mg/L) is

$$C = \begin{pmatrix} 1.2 & 0.4 & 2.1 \\ 0.9 & 1.5 & 0.3 \\ 2.0 & 0.7 & 1.1 \end{pmatrix}$$

- (a) Compute the trace of C (sum of diagonal entries).
- (b) Compute $\det(C)$.
- (c) Suppose the safe-limit vector is $b = \begin{pmatrix} 1.0 \\ 1.0 \\ 1.5 \end{pmatrix}$ and each row of C represents a sample. Write a linear system $C^T x = b$ and solve it (if possible) using Cramer's rule. Interpret the result.

— End of Unit 2 Notes —